

DEEP LEARNING ARCHITECTURES — PURE OBJECT-ORIENTED (OOP) PIPELINE FLOWCHART

Batch 2 — Week 2 Submission | Author: Sardar Ahmed | Institution: Alpatron Technologies

MASTER PIPELINE ORCHESTRATOR: class DeepLearningPipeline (Main.py)

Attributes: +ann_step +cnn_step +rnn_step +lstm_step +comparator_step
Method: +run() -> Coordinates end-to-end execution, tracks execution time, & logs metrics

DOMAIN 1: COMPUTER VISION (MNIST Digit Recognition — 28x28 Images)

STEP 1: ANN (MLP) class ArtificialNeuralNetwork

Dataset: MNIST (784 features)

- +load_data()
- +preprocess_data() (/255.0)
- +build_model() (Dense 128->64)
- +compile_model() (Adam, CE)
- +train() (10 epochs, 32 batch)
- +evaluate() (97.30% Acc)
- +predict(samples)
- +plot_training_history()
- +plot_sample_predictions()
- +save_model() (.keras)
- +run() -> dict

STEP 2: CNN (2D Conv) class ConvolutionalNeuralNetwork

Dataset: MNIST (28x28x1)

- +load_data()
- +preprocess_data() (Reshape)
- +build_model() (Conv->Pool->D0)
- +compile_model() (Adam, CE)
- +train() (10 epochs, 64 batch)
- +evaluate() (99.22% Acc)
- +predict(samples)
- +plot_training_history()
- +plot_sample_predictions()
- +save_model() (.keras)
- +run() -> dict

DOMAIN 2: NATURAL LANGUAGE PROCESSING (IMDB Sentiment — 200 Tokens)

STEP 3: SimpleRNN class RecurrentNeuralNetwork

Dataset: IMDB (10K Vocab)

- +load_data()
- +preprocess_data() (Pad 200)
- +build_model() (Embed->RNN)
- +compile_model() (Adam, BCE)
- +train() (5 epochs, 64 batch)
- +evaluate() (72.16% Acc)
- +predict(samples)
- +decode_review(tokens)
- +plot_training_history()
- +save_model() (.keras)
- +run() -> dict

STEP 4: LSTM Network class LSTMNeuralNetwork

Dataset: IMDB (10K Vocab)

- +load_data()
- +preprocess_data() (Pad 200)
- +build_model() (Embed->LSTM)
- +compile_model() (Adam, BCE)
- +train() (5 epochs, 64 batch)
- +evaluate() (84.85% Acc)
- +predict(samples)
- +decode_review(tokens)
- +plot_training_history()
- +save_model() (.keras)
- +run() -> dict

STEP 5: CROSS-ARCHITECTURE BENCHMARKING (class DLModelComparator)

- Evaluates saved models from disk & captures in-memory metrics (ANN: 97.30%, CNN: 99.22%, RNN: 72.16%, LSTM: 84.85%)
 - Exports Metrics Table: ModelComparison/deep_learning_benchmark_results.csv
- Generates Comparison Charts: vision_models_comparison.png | sequence_models_comparison.png | overall_dl_benchmark_comparison.png